

COVID-19 Impact & Vaccination Effectiveness

Full Analysis Documentation

Dataset: Our World in Data (OWID)
405,726 records · 243 countries · Jan 2020 – Aug 2024

Tools: SQL Server (T-SQL) · Microsoft Excel

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1. Executive Summary

This project analyses the global COVID-19 pandemic using the Our World in Data daily dataset, covering 405,726 rows across 243 countries and territories from January 2020 to August 2024. The analysis is structured in six SQL phases, from data import to confounder analysis, and surfaces three primary findings:

- **Vaccination reduced death rates by 2.5×** — countries above 80% vaccination averaged a 0.57% case-fatality rate vs. 1.40% for countries below 20%.
- **GDP per capita is a stronger predictor than continent** — a smooth 5.5× death-rate gradient from the poorest to richest country bands, vs. noisy continent-level rollups.
- **Africa's low reported mortality is a data artefact** — 182 deaths per million vs. 2,581 in Europe almost certainly reflects under-testing and under-reporting, not genuine immunity.

Global Dataset Snapshot

Metric	Value
Total Cases	775.9 Million
Total Deaths	7.06 Million
Countries / Territories	243
Date Range	Jan 2020 – Aug 2024
Avg Country Death Rate (CFR)	1.23%
Avg Country Vaccination Rate	58.0%

2. Methodology

The analysis follows a six-phase pipeline executed entirely in SQL Server (T-SQL), with results exported to Excel for visualisation.

Phase 0 — Data Import & Exploration

Loaded the OWID CSV into `dbo.OwidCovid`. Verified row count (405,726), date span, and per-continent country coverage. Identified region aggregates (World, European Union, High income, etc.) with continent IS NULL for exclusion.

Phase 1 — Country_Summary View

Created `dbo.Country_Summary` — one row per country using the latest available date. Computed three derived rates: `InfectionRate`, `DeathRate (CFR)`, and `VaccinationRate` using `NULLIF` to guard against division-by-zero.

Phase 2 — Data Quality Checks

Checked missing values across four critical columns. Confirmed zero duplicate (location, date) pairs. Flagged ~2,800 rows with negative new_cases (data revisions from national authorities) — treated as noise, not removed.

Phase 3 — Trend & Seasonality

Aggregated global monthly new cases and deaths (56-month series). Computed monthly CFR. Built a 7-day rolling average window for the 12 most-populous countries using ROWS BETWEEN 6 PRECEDING AND CURRENT ROW.

Phase 4 — Vaccination Effectiveness

Bucketed Country_Summary rows into five vaccination bands (0-20%, 20-40%, 40-60%, 60-80%, 80%+). Compared average, min, and max DeathRate per band. This is the primary analytical finding.

Phase 5 — GDP & Continent Confounders

Repeated the banding logic for GDP per capita (\$5k increments). Overlaid average VaccinationRate per GDP band to test co-linearity. Added a continent-level SUM rollup with deaths-per-million normalisation.

3. Key Findings

3.1 Vaccination vs. Death Rate

Grouping 243 countries into five vaccination bands and comparing average CFR reveals a clear monotonic relationship — with one anomaly in the 20-40% band driven by high-mortality outliers with early reporting gaps.

Vaccination Band	Countries	Avg Death Rate	Signal
0 – 20%	40	1.40%	High mortality
20 – 40%	36	2.04%	Outlier-inflated
40 – 60%	50	1.35%	Mid range
60 – 80%	72	0.88%	Improving
80%+	45	0.57%	Lowest mortality

→ The 80%+ band has a 2.5× lower death rate than the 0-20% band (0.57% vs 1.40%) across 85 countries in the extreme cohorts.

3.2 GDP Per Capita — The Stronger Predictor

GDP per capita produces a smoother, monotonically decreasing death-rate gradient than any continent grouping. Vaccination rate tracks GDP almost in lockstep, confirming that access — not hesitancy — drove the protection gap.

GDP Band	Avg Death Rate	Avg Vax Rate	Interpretation
< \$5k	1.92%	34%	Highest risk
\$5k – \$15k	1.61%	52%	Elevated
\$15k – \$30k	1.14%	60%	Moderate
\$30k – \$50k	0.58%	73%	Low risk
\$50k+	0.35%	79%	Lowest risk

→ A 5.5× gap in death rate between <\$5k and \$50k+ GDP bands. The 2.3× gap in vaccination rate maps directly onto this mortality gradient.

3.3 Deaths Per Million by Continent

Continent	Deaths / Million	Note
South America	3,104	Highest — limited early ICU capacity
North America	2,784	US drives the figure

Europe	2,581	Second wave hit elderly populations hard
Oceania	731	Delayed arrival; vaccination lead helped
Asia	347	Heterogeneous; India/Indonesia weigh heavily
Africa	182	■ Likely severe undercounting

3.4 Timeline & CFR Evolution

- **Mar–Apr 2020:** CFR reported at 6-9%. Testing was limited to severe cases; true prevalence was vastly undercounted.
- **Late 2020 (Alpha wave):** First major global case peak. Monthly deaths peaked Dec 2020 – Jan 2021.
- **Dec 2021 – Feb 2022 (Omicron):** Case volume ~4× the Alpha peak, but monthly CFR had fallen below 0.3% as vaccine rollout shifted the population's immunity profile.
- **Mid-2022 onward:** Monthly CFR stabilised below 0.5%, with periodic upticks as new variants evaded partial immunity.

4. Country Rankings

Top 10 Countries by Total Deaths

Country	Total Deaths	CFR
United States	1,193,165	1.15%
Brazil	702,116	1.87%
India	533,623	1.18%
Russia	403,188	1.66%
Mexico	334,551	4.39%
United Kingdom	232,112	0.93%
Peru	220,975	4.88%
Italy	197,542	0.74%
Germany	174,979	0.46%
France	167,985	0.43%

Top 10 Countries by Vaccination Rate

Country	Vax Rate	CFR
Qatar	105.8%	0.13%
United Arab Emirates	103.7%	0.22%
Chile	90.3%	1.19%
China	89.5%	0.12%
Spain	85.7%	0.87%
Peru	84.3%	4.88%
Japan	83.4%	0.22%
Canada	82.6%	1.15%
Brazil	81.8%	1.87%
Portugal	81.2%	0.51%

Note: Peru demonstrates that high vaccination does not guarantee low CFR when early-pandemic deaths (before vaccines arrived) dominate the cumulative total.

5. Analytical Insights

✓ Vaccination materially lowered death rates

The 2.5x spread between extreme vaccination bands is consistent across multiple analytical cuts. The effect holds after controlling for GDP, suggesting vaccines provided independent protection beyond the wealth-access effect.

i GDP is the stronger predictor than continent

Continent rollups hide wide within-continent variation. GDP per capita bands produce a smooth monotonic gradient in both death rate and vaccination rate, making it the cleaner analytical dimension.

! Africa's low deaths-per-million is an artefact

182 deaths/M vs. Europe's 2,581 is implausible given Africa's weaker healthcare infrastructure. Low testing capacity and fragmented reporting systems mean the true toll is almost certainly far higher.

i Two clear waves dominate the timeline

Alpha (late 2020) and Omicron (winter 2021-22) are the two dominant case peaks. Omicron was ~4x larger by case volume but with a CFR roughly 10x lower - the population's immunity profile had fundamentally shifted.

! Early-pandemic CFR was wildly inflated

March-April 2020 CFRs of 6-9% reflected near-zero community testing. By mid-2022 the monthly figure was below 0.5%. Most of this is measurement, not biology - a critical caveat for any cross-country CFR comparison.

x Income gap drove the vaccination gap

<\$5k GDP countries averaged 34% vaccination vs. 79% for \$50k+ countries. The 2.3x protection gap maps directly onto a 5.5x mortality gap - supply and logistics, not policy alone, determined who survived.

6. Recommendations

[HIGH] Equity in vaccine distribution

Low-income countries averaged 34% vaccination vs. 79% for high-income nations. Global supply mechanisms (COVAX-style) require stronger early-pandemic funding and faster delivery timelines - the current model reacted too slowly.

[HIGH] Invest in surveillance infrastructure

Undercounting in low-resource regions distorts the global picture and delays outbreak detection. Routine population testing and seroprevalence studies should be standing public-health infrastructure, not crisis-mode add-ons.

[MED] Standardise outcome reporting

CFR comparisons are unreliable without consistent definitions for COVID-attributed death, testing rate normalisation, and population denominators. WHO-coordinated reporting standards would make future cross-country analysis trustworthy.

[LOW] Preserve the data legacy

OWID's daily country-level dataset is one of the few clean, comparable cross-country pandemic archives. Funding open, longitudinal datasets is the cheapest preparedness investment a government can make.

7. SQL Phase Reference

The complete SQL code is in `sql/covid_analysis.sql`. Key snippets are reproduced below for reference.

Country_Summary View (Phase 1)

```
CREATE VIEW dbo.Country_Summary AS
WITH LatestPerCountry AS (
  SELECT location, continent, MAX(date) AS LatestDate
  FROM dbo.OwidCovid WHERE continent IS NOT NULL
  GROUP BY location, continent
)
SELECT o.location AS Country, o.continent AS Continent,
  CAST(o.total_deaths AS FLOAT) / NULLIF(o.total_cases, 0) AS DeathRate,
  CAST(o.people_fully_vaccinated AS FLOAT) / NULLIF(o.population,0) AS VaccinationRate
FROM dbo.OwidCovid o
INNER JOIN LatestPerCountry lpc ON o.location=lpc.location AND o.date=lpc.LatestDate;
```

Vaccination Band Analysis (Phase 4 — key query)

```
WITH Banded AS (
  SELECT Country, DeathRate,
  CASE WHEN VaccinationRate < 0.20 THEN '1. 0-20%'
  WHEN VaccinationRate < 0.40 THEN '2. 20-40%'

```



```
WHEN VaccinationRate < 0.60 THEN '3. 40-60%'
WHEN VaccinationRate < 0.80 THEN '4. 60-80%'
ELSE '5. 80%+' END AS VaxBand
FROM dbo.Country_Summary
WHERE VaccinationRate IS NOT NULL AND DeathRate IS NOT NULL
)
SELECT VaxBand, COUNT(*) AS Countries, AVG(DeathRate) AS AvgDeathRate
FROM Banded GROUP BY VaxBand ORDER BY VaxBand;
```